

Mini Plants vs Zombies Game - Planning Sheet

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Section: Lithium

Date: October 1, 2026

Subject: Grade 9 Computer Science 3 | First Quarter | Pair Activity | Alternative Assessment

1. Before Coding (Computational Thinking)

Decomposition

To make the Mini Plants vs Zombies game work, we break it down into these smaller components:

Plant Class Definition: Model attributes (name, health, damage) and methods (attack, take_damage).

Zombie Class Definition: Model attributes (name, health, damage, distance) and methods (move, attack, take_damage).

Object Initialization: Instantiating two unique Plant objects (e.g., Peashooter and Snow Pea) and one Zombie object.

Game Loop: Managing sequential plant attacks, checking zombie movement vs. melee attack conditions, shifting targets when a plant falls, and checking win/loss conditions.

Output Display: Printing clear status messages after each turn showing updated HP, distances, and action details.

Pattern Recognition

The following actions and checks repeat across multiple turns in the game:

Turn Check: Each turn checks standard conditions in a fixed sequence until a win or loss trigger occurs.

Consecutive Plant Attacks: Each alive plant performs its attack(zombie) method once per turn in order.

Zombie Action Decision: Checking if zombie.distance > 0: the zombie performs move(); else: the zombie performs attack(current_target_plant).

Target Switching Pattern: If plant1.health <= 0, the zombie automatically targets plant2.

Health & Termination Check: Evaluating whether zombie.health <= 0 (Plants Win) or both plants' health <= 0 (Zombie Wins) at the end of each sequence.

Abstraction

To keep the game focused on essential interactions, we include only necessary attributes and actions while omitting complex details:

Included Details:

name: Identifies the specific plant or zombie.

health: Tracks remaining life points.

damage: Quantitative attack power of each unit.

distance: Tracks the single-lane step distance between the zombie and the lead

plant.

attack() & take_damage(): Core computational interaction between objects.

move(): Reduces zombie distance step-by-step.

Algorithm Design

The step-by-step game execution flow is structured as follows:

Initialize Objects:

Create plant1 (e.g., Peashooter, HP: 30, Damage: 15).

Create plant2 (e.g., Snow Pea, HP: 20, Damage: 10).

Create zombies (e.g., Zombie, HP: 70, Damage: 15, Distance: 3).

Set Initial Target: Set current target plant = plant1.

Start Game Loop: Continue looping while zombie.health > 0 and (plant1.health > 0 or plant2.health > 0):

Step 3a (Plant 1 Turn): If plant1.health > 0, call plant1.attack(zombie).

Check: If zombie.health <= 0, break loop immediately.

Step 3b (Plant 2 Turn): If plant2.health > 0, call plant2.attack(zombie).

Check: If zombie.health <= 0, break loop immediately.

Step 3c (Zombie Turn):

Determine active target: If plant1.health <= 0, set target = plant2.

If zombie.distance > 0: Call zombie.move() (reduces distance by 1).

Else (zombie.distance == 0): Call zombie.attack(target).

Step 3d (Display Status): Print current turn recap with health values and zombie distance.

Determine and Print Game Outcome:

If zombie.health <= 0: Print "Victory! The plants defeated the zombie!"

Else if plant1.health <= 0 and plant2.health <= 0: Print "Game Over! The zombie ate your brains!"

2. Work Plan

Session	Time	What to Do\do
1	10 minutes	Plan the game using the four thinking skills.
1	10 minutes	Draw the UML diagram and show the association.
1	25 minutes	Code the two classes, create three objects, and try one plant attacking the Zombie.
1	5 minutes	Show the teacher both plants, the Zombie, and one working interaction.
2	25 minutes	Finish the turn loop, two plant attacks, movement, target selection, and winner.
2	10 minutes	Run and record the test cases below; correct any errors.
2	15 minutes	Demonstrate the game. Both partners explain part of the code.

3. Testing Checklist

- ☐ The two plants have different damage values
- ☐ Plants defeat the Zombie:
- ☐ Zombie reaches distance 0 and damages a plant:
- ☐ Zombie targets the second plant after the first is defeated:
- ☐ Game stops when the Zombie or both plants are defeated:

